

Class Activity-2-MC

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Course Code: MLA0305

Course Name: Reinforcement learning

R code:

```
# =====  
  
# Monte Carlo Process - Summary + Flowchart + Visualization  
# Real-Time Example: Student Exam Pass/Fail Simulation  
# =====  
  
# Install package if not available  
if (!require(DiagrammeR)) install.packages("DiagrammeR")  
library(DiagrammeR)  
  
set.seed(123)  
  
# -----  
# STEP 1: Generate Episodes  
# -----  
episodes <- 1000  
  
# Probability of passing = 70%  
results <- sample(c("Pass", "Fail"),  
                 size = episodes,  
                 replace = TRUE,  
                 prob = c(0.7, 0.3))  
  
# -----  
# STEP 2: Collect Rewards
```

```

# Pass = Reward 1
# Fail = Reward 0
# -----
reward <- ifelse(results == "Pass", 1, 0)

# -----
# STEP 3: Compute Return
# Average Reward
# -----
average_return <- mean(reward)

# -----
# STEP 4: Update Values
# Pass/Fail Counts
# -----
pass_count <- sum(results == "Pass")
fail_count <- sum(results == "Fail")

# -----
# STEP 5: Summary
# -----
cat("=====\n")
cat("  MONTE CARLO SIMULATION SUMMARY\n")
cat("=====\n")
cat("Episodes      :", episodes, "\n")
cat("Passed Students  :", pass_count, "\n")
cat("Failed Students  :", fail_count, "\n")
cat("Average Reward   :", round(average_return,3), "\n")
cat("Pass Percentage  :", round(pass_count/episodes*100,2), "%\n")
cat("Fail Percentage   :", round(fail_count/episodes*100,2), "%\n")

```

```

# =====

# FLOWCHART

# =====

grViz("
digraph MonteCarlo {

graph [layout = dot, rankdir = TB]

node [
shape = rectangle,
style = filled,
fillcolor = lightblue,
fontname = Helvetica,
fontsize = 14
]

A [label='Episode']
B [label='Collect Rewards']
C [label='Compute Return']
D [label='Update Values']
E [label='Repeat']

A -> B
B -> C
C -> D
D -> E
E -> A
}

```

")

```
# =====
```

```
# BAR CHART
```

```
# =====
```

```
counts <- c(pass_count, fail_count)
```

```
barplot(counts,  
  names.arg=c("Pass","Fail"),  
  col=c("blue","pink"),  
  main="Monte Carlo Simulation Result",  
  ylab="Number of Students")
```

```
text(c(0.3,1),  
  counts+20,  
  labels=counts)
```

```
# =====
```

```
# PIE CHART
```

```
# =====
```

```
pie(counts,  
  labels=paste(c("Pass","Fail"),  
    "\n",  
    round(counts/sum(counts)*100,1,"%"),  
  col=c("green","red"),  
  main="Pass vs Fail Distribution")
```

```
# =====
```

```
# TABLE
```

```
# =====
```

```
summary_table <- data.frame(
```

```

Outcome=c("Pass","Fail"),

Count=counts,

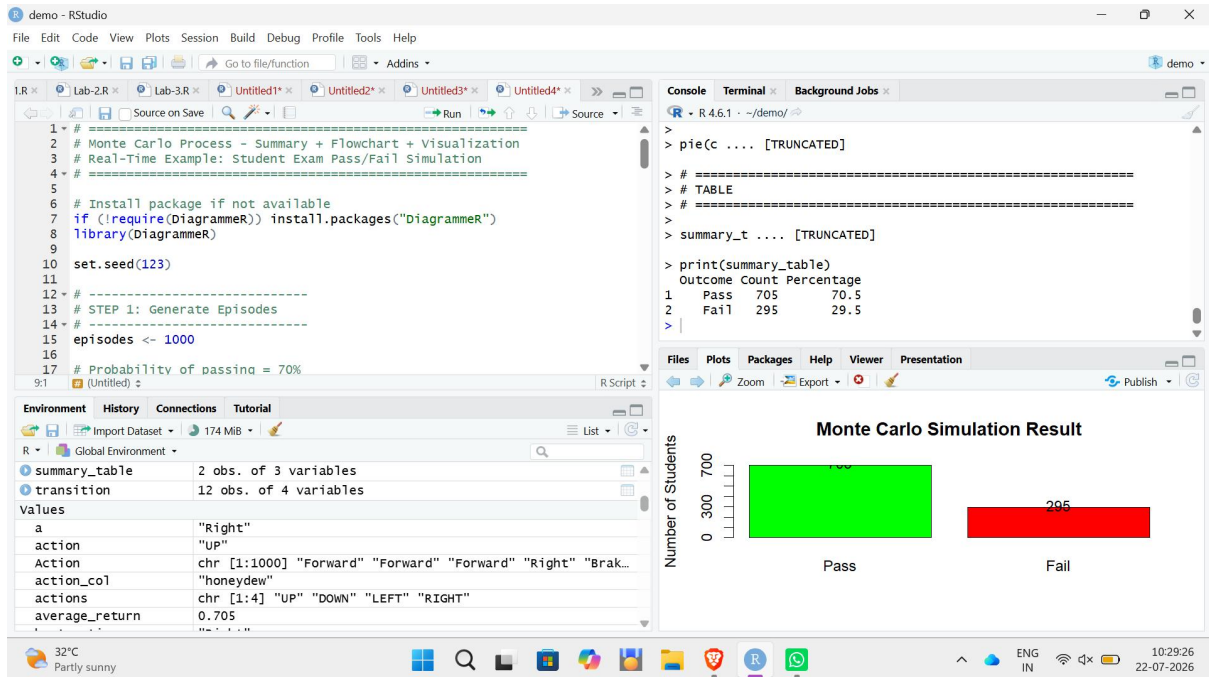
Percentage=round(counts/sum(counts)*100,2)

)

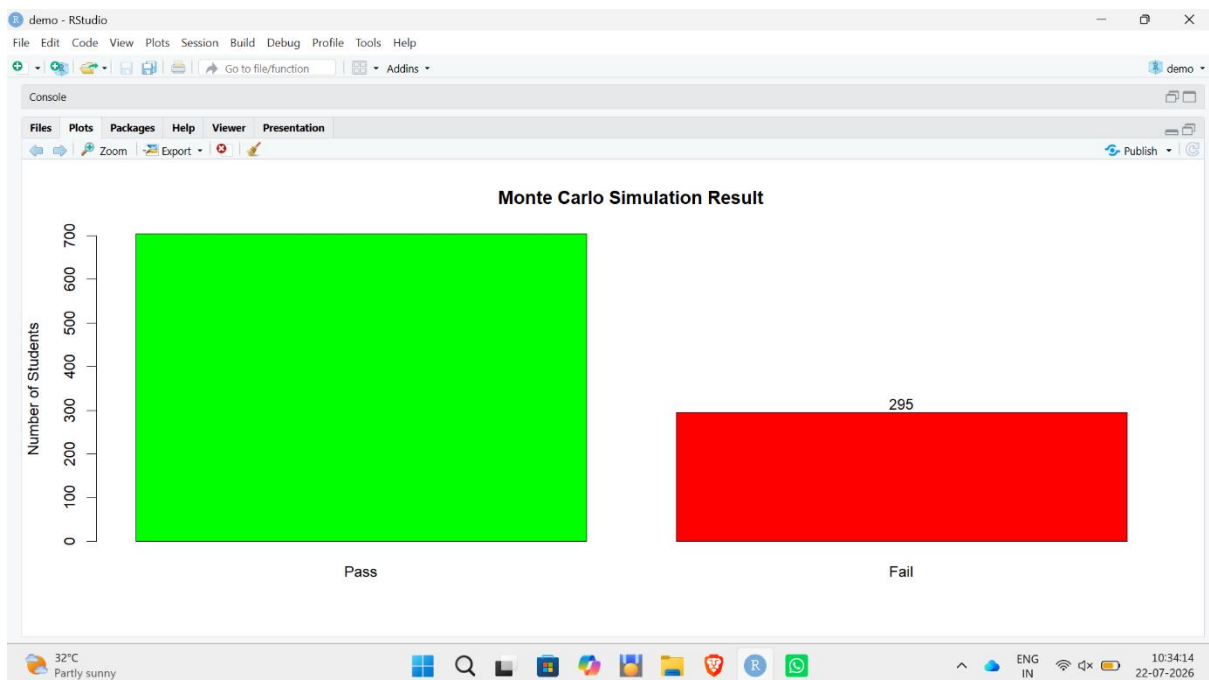
print(summary_table)

```

Full screenshot of Rstudio:



Visual result:



Summary result:

```
demo - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Console Terminal Background Jobs
R - R 4.6.1 - ~/demo/
> # =====
> count .... [TRUNCATED]
> barplot(counts,
+   names.arg=c("Pass","Fail"),
+   col=c("green","red"),
+   main="Monte Carlo Simulation Result",
+   ylab .... [TRUNCATED]
> text(c(0.3,1),
+   counts+20,
+   labels=counts)
> # =====
> # PIE CHART
> # =====
> pie(c .... [TRUNCATED]
> # =====
> # TABLE
> # =====
> summary_t .... [TRUNCATED]
> print(summary_table)
Outcome Count Percentage
1 Pass 705 70.5
2 Fail 295 29.5
> |
Files Plots Packages Help Viewer Presentation
32°C Partly sunny
10:33:48 22-07-2026
```